Lesson 10 Adjust Bus Servo Speed

This lesson aims to control the rotation speed of the bus servo on the expansion board. For detailed introduction of bus servo, please refer to "4.Expanded Lesson\3.Raspberry Pi Expansion Board Lesson\2.Expansion Board Control\Lesson 8 Read Bus Servo Status".

1. Working Principle

In the program, modify the ID, runtime, and position of the servo to control the rotation of the bus servo. The source code of the program is located in "/home/pi/board_demo/bus_servo_speed.py".

```
# 关闭前处理(precess before closing)
def Stop(signum, frame):
    global start
    start = False
    print('关闭中...(closing...)')

signal.signal(signal.SIGINT, Stop)

if __name__ == '__main__':
    while True:
        board.bus_servo_set_position(0.5, [[1, 0],[2, 0]])
        time.sleep(0.5)
        board.bus_servo_set_position(2, [[1, 1000], [2, 1000]])
        time.sleep(1)
        board.bus_servo_stop([1, 2])
        time.sleep(1)
        if not start:
            board.bus_servo_stop([1, 2])
            time.sleep(1)
            print('已关闭(closed)')
            break
```

The program calls the "board.bus_servo_set_position" function in the "Board" library to implement the servo control.

①The "board.bus_servo_set_position" function is used to set the angle position of the bus servo. It receives two parameters: "0.5" represents the time of the servo rotation; "[[1, 0], [2, 0]]" represents the servo and its corresponding angle position to be set.

② "[[1, 0], [2, 0]]" is used to set the angle positions of servos 1 and 2 to "0" degree.

③ Use the "time.sleep" function to wait for 0.5s to ensure that the servo has enough time to rotate to the specified angle.

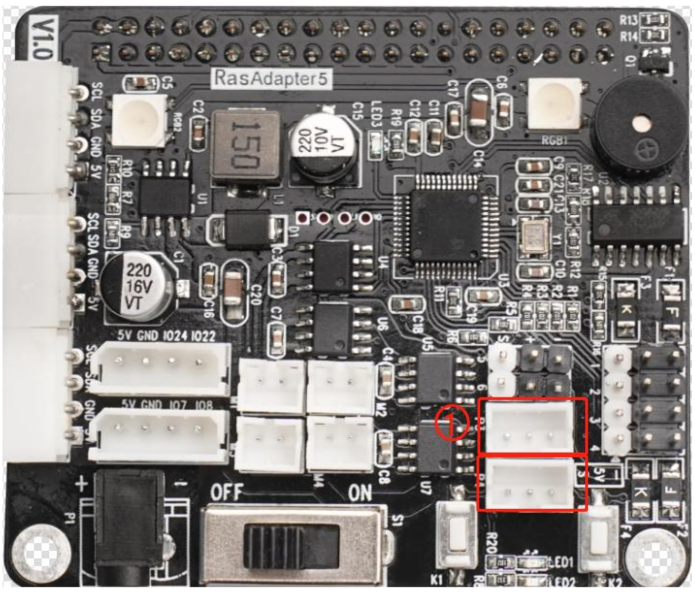
④ Then, the "board.bus_servo_set_position" is called again to set the angle positions of servos 1 and 2 to 1000 degrees. The first parameter "2" indicates the rotation time is 2.

⑤ Similarly, use the "board.bus_servo_stop" function to wait for 1s to rotate the servo to the specified angle.

⑥ The "board.bus_servo_stop" function is called to stop the motion of servos 1 and 2. It takes a parameter, which is a list of servo numbers to be stopped. In this case, passing "[1, 2]" as the parameter means stopping the motion of servos 1 and 2.

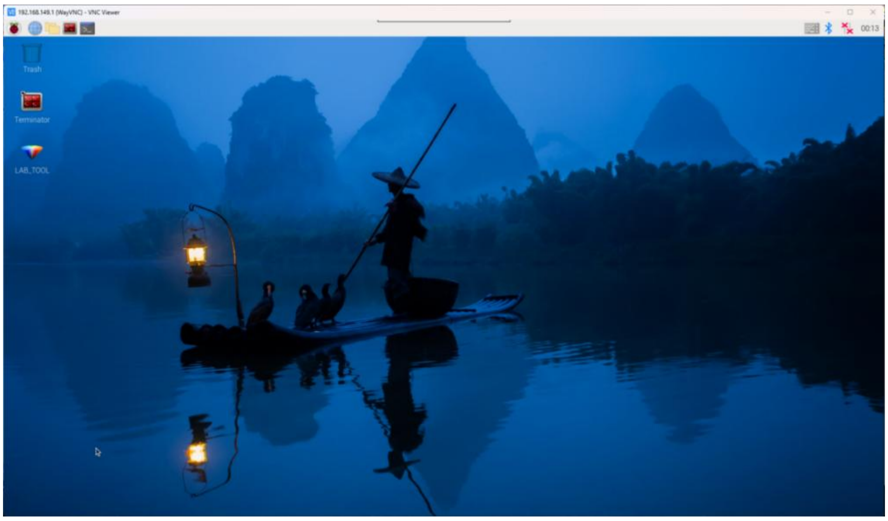
2. Getting Ready

There are 2 bus servo interfaces on the Raspberry Pi expansion board. Let's test by connecting the No.1 bus servo interfaces, as shown below:



3. Operation Steps

1) Turn the device on and connect the robot through the VNC remote connection tool.



2) Click  in the top left corner of the desktop, or press the shortcut key “Ctrl+Alt+T” to open the command line terminal.

3) Enter the command “cd board_demo/” to navigate to the directory where the demo program is located, then press “Enter”.

Enter the command of running the program “python3 bus_servo_ speed.py”, and press “Enter”.

```
pi@raspberrypi:~/board_demo $ python3 bus_servo_speed.py

**********************************************************
********功能:幻尔科技树莓派扩展板，控制总线舵机转动(function:Hiwonder Raspberry
Pi expansion board, bus servo rotation control)**********
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Official website:https://www.hiwonder.com
Online mall:https://hiwonder.tmall.com
----------------------------------------------------------
Tips:
 * 按下Ctrl+C可关闭此次程序运行，若失败请多次尝试！(Press "Ctrl+C" to exit the p
rogram. If it fails, please try again multiple times!)
----------------------------------------------------------
```

4) To close the program, press “Ctrl+c”. If the program cannot be closed successfully, repeat this operation until it exits.

4. Program Outcome

Once the program is executed, the Raspberry Pi expansion board sets the servo to the initial position (0), then rotates it to the specified angle position (1000), and finally stops its motion. By adjusting the parameters, the Raspberry Pi expansion board can control the angle position and runtime of different servos. This allows the speed control of the servo.